



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

FashionPulse AI — Sales Trends, Customer Behavior & Predictive Intelligence

The domain of the Project: Data Analytics & Artificial Intelligence

COURSE NAME: DATA ANALYTICS

Team Mentors

(and their designation):

Keerthana V. Ma'am (Senior Executive - Business Analyst Exl Service)

Team Member:

Mr. SATYAM KUMAR SHRIVASTAV - G2 Data Analytics

Period of the project

July 2026 to September 2026



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Declaration

The project titled “**FashionPulse AI — Sales Trends, Customer Behavior & Predictive Intelligence**” has been mentored by **KeerthanaV. Ma’am**, organised by SURE Trust, from July 2026 to September 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Mr. Satyam Kumar Shrivastav

Mentor’s Name

Keerthana V.

Senior Executive

Business Analyst Exl Service

Prof. Radhakumari

Executive Director & Founder

SURE Trust



Table of contents

Chapter No.	Content	Page No.
1	Executive Summary	4
2	Introduction	5
3	Project Objectives	6
4	Methodology and Results	13
5	Social/Industry Relevance of the Project	16
6	Learning and Reflection	17
7	Conclusion and Future Scope	18
8	Project Structure and Technical Reference	19



1. Executive Summary

FashionPulse AI is an end-to-end business intelligence and predictive analytics solution designed for a women's fashion e-commerce catalog. The project combines a Flask REST API, MySQL data store, SQL-based analytics, an interactive browser dashboard, and four optional machine-learning workflows to transform product-level data into actionable business intelligence.

The system helps an e-commerce or merchandising team understand catalog scale, sales performance, product attributes, pricing and discount relationships, inventory health, returns, and predictive product outcomes. The dashboard contains five major views: Executive Overview, Product Performance, Pricing Intelligence, Inventory and Returns, and AI Predictions.

The supplied dataset contains 50,000 product rows. Product attributes include category, subcategory, material, color, size, season, style, price, discount, rating, review count, stock quantity, units sold, and return rate. The data is loaded into MySQL and exposed through Flask APIs for dashboard consumption.

The predictive layer contains four workflows: sales regression, performance classification, return-risk prediction, and product segmentation. The sales predictor compares Ridge, Random Forest, Gradient Boosting, and optionally XGBoost and LightGBM using holdout R-squared. The performance classifier uses sales tertiles and compares models using weighted F1. The return-risk model identifies products at or above the 75th percentile of return rate as high risk. K-Means clustering creates four configured product segments.

Overall, FashionPulse AI demonstrates how data engineering, SQL analytics, visualization, machine learning, and web application development can be integrated into one business-focused analytical platform.



2. Introduction

2.1 Background and Context of the Project

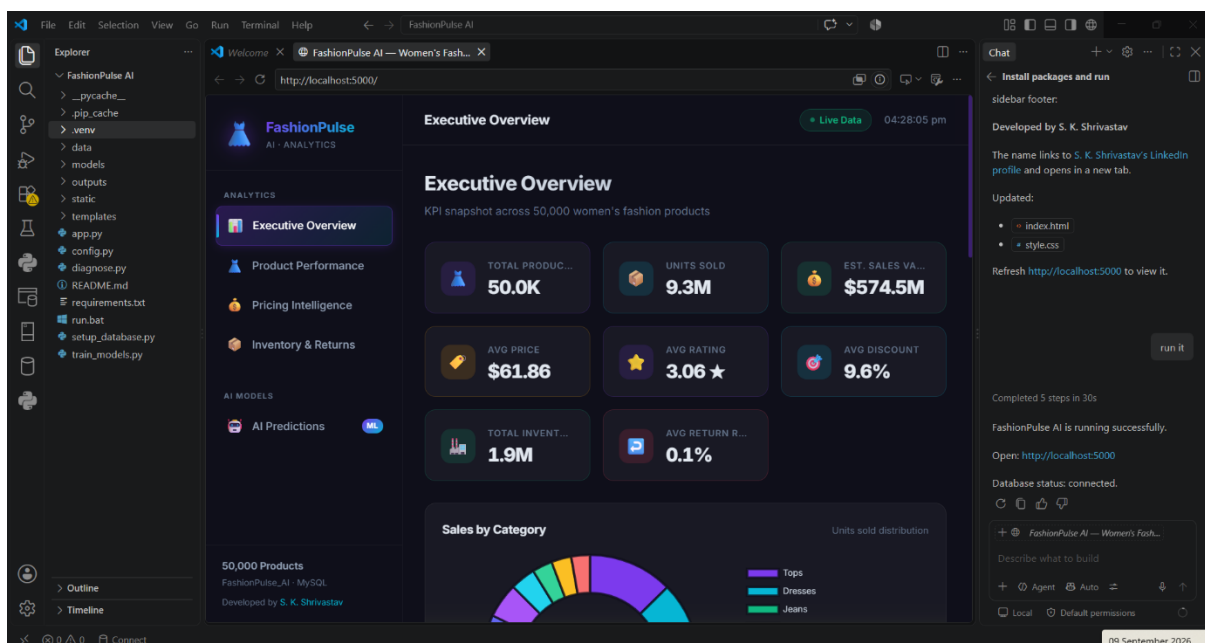
Fashion e-commerce businesses generate large volumes of product and sales-related information. Understanding which categories, styles, seasons, price bands, discounts, and products perform well is important for merchandising and inventory decisions. Raw tabular data is difficult to interpret directly, particularly when users need to combine multiple dimensions of product performance.

2.2 Problem Statement

The main problem addressed by the project is how to transform product-level e-commerce data into a reliable, interactive, and predictive decision-support system that helps users understand sales performance, pricing, inventory, returns, and likely product outcomes.

2.3 Scope of the Project

The project covers product catalog analytics, sales performance, pricing intelligence, inventory health, return analysis, interactive dashboard visualization, and four machine-learning workflows. The current implementation is based on a 50,000-row women's fashion catalog dataset.





3. Project Objectives

3.1 Objectives of the Project

The primary objective is to develop an integrated business intelligence and predictive analytics platform for a women's fashion e-commerce catalog.

1. Load and manage the 50,000-row catalog dataset in MySQL.
2. Provide SQL-based analytical views for products, sales, categories, seasons, styles, pricing, inventory, and returns.
3. Develop an interactive dashboard that makes business metrics accessible without direct database querying.
4. Analyze pricing and discount relationships with sales performance.
5. Identify inventory health and products requiring restocking attention.
6. Build machine-learning workflows for sales prediction, performance classification, return-risk scoring, and product segmentation.
7. Expose analytics and prediction capabilities through Flask REST APIs.
8. Demonstrate practical skills across Python, MySQL, SQL analytics, JavaScript dashboard development, and machine learning.

3.2 GitHub & Overview of the Project

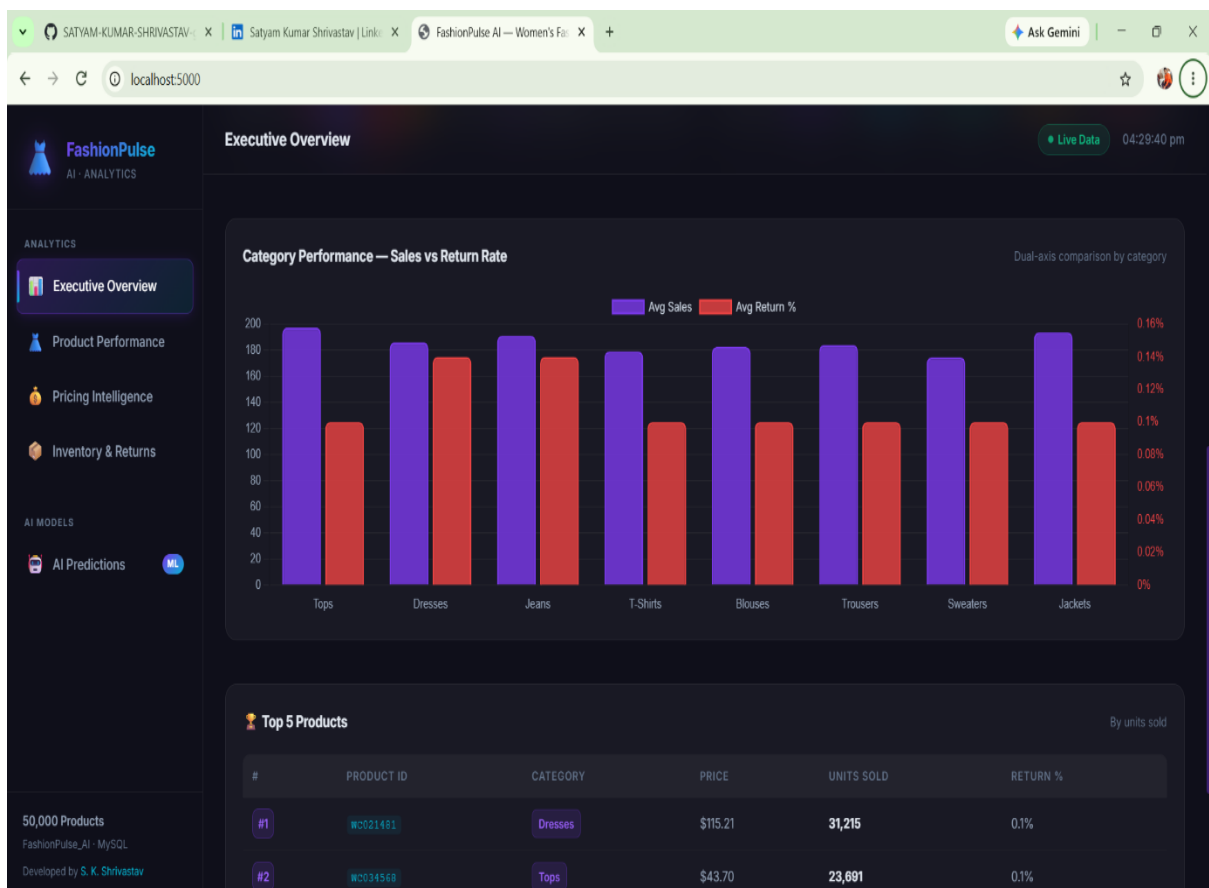
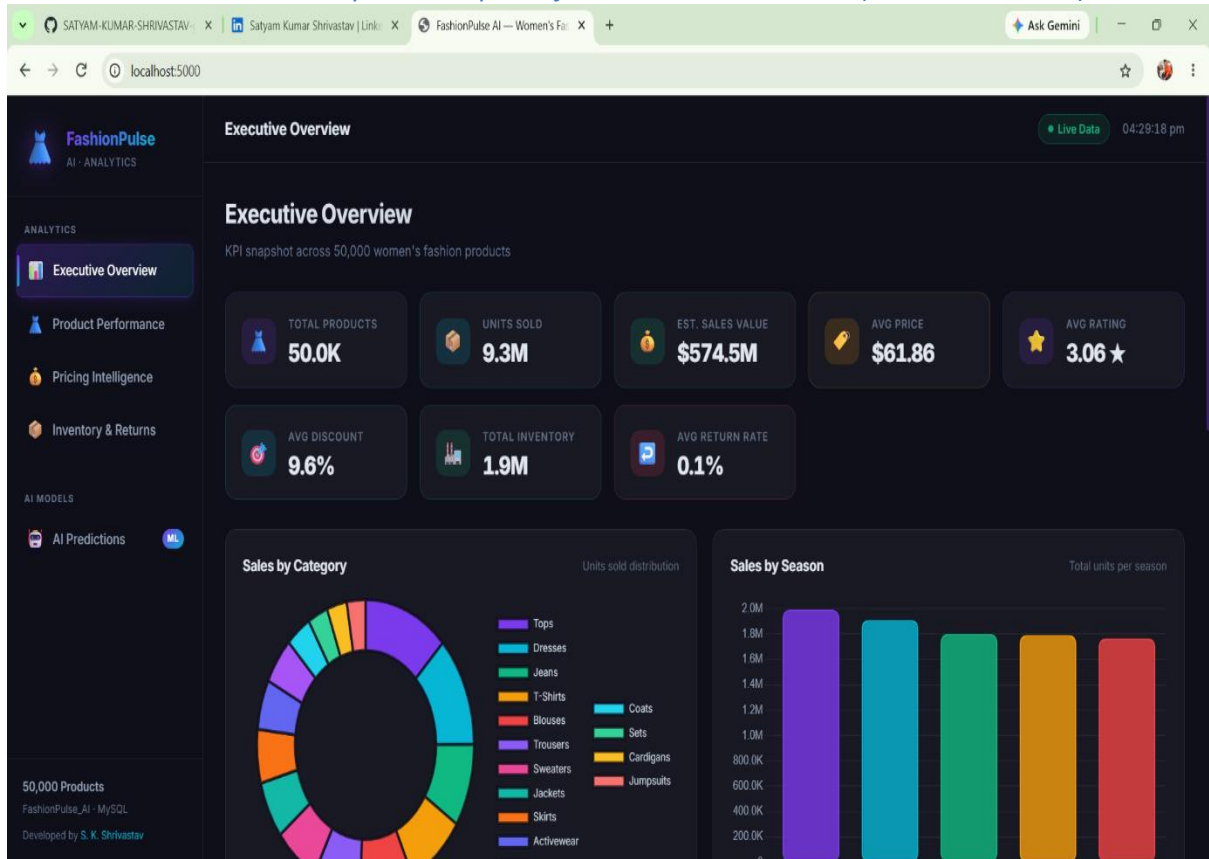
GitHub: <https://github.com/sure-trust/SATYAM-KUMAR-SHRIVASTAV-g2-data-analytics/tree/main/Final%20capstone%20project>

Project Overview:

Link: <http://localhost:5000/>

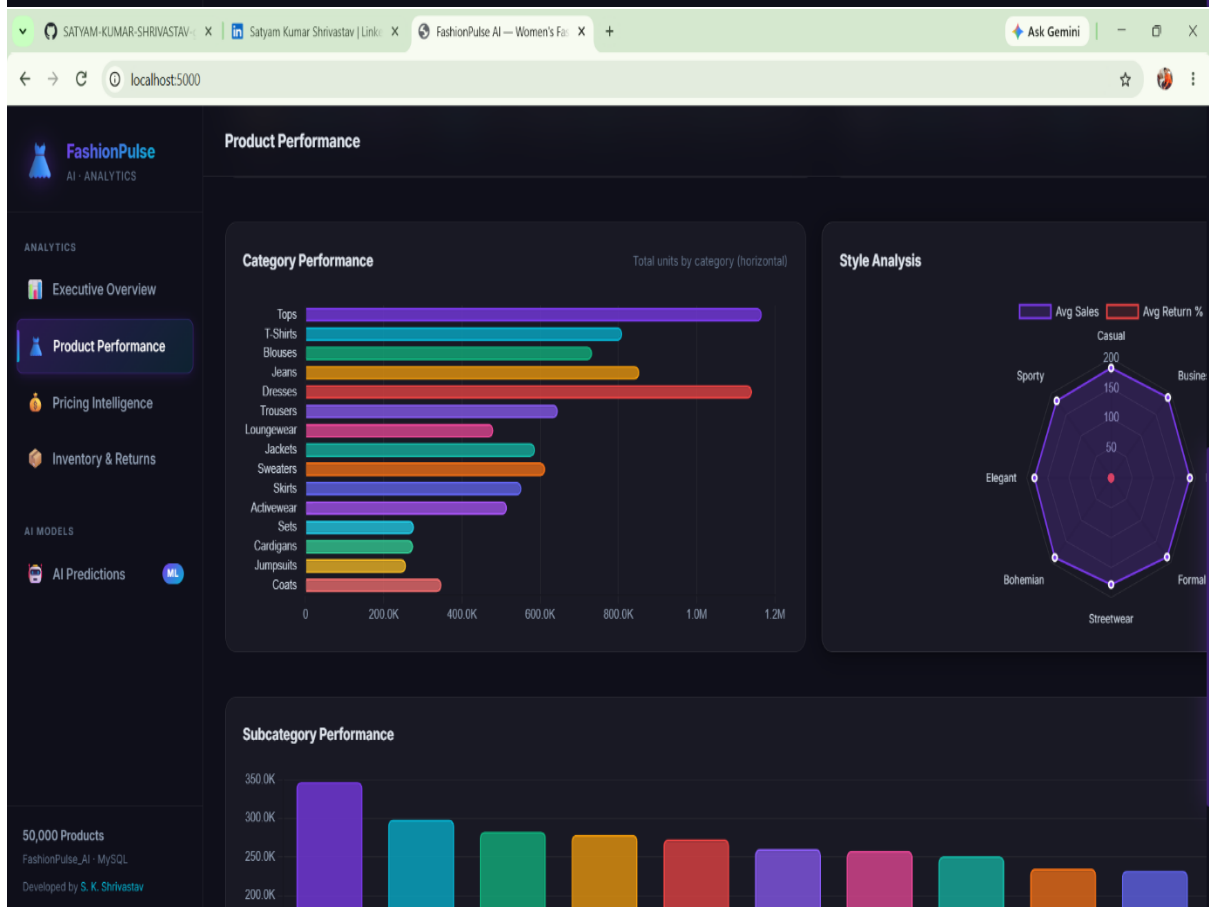
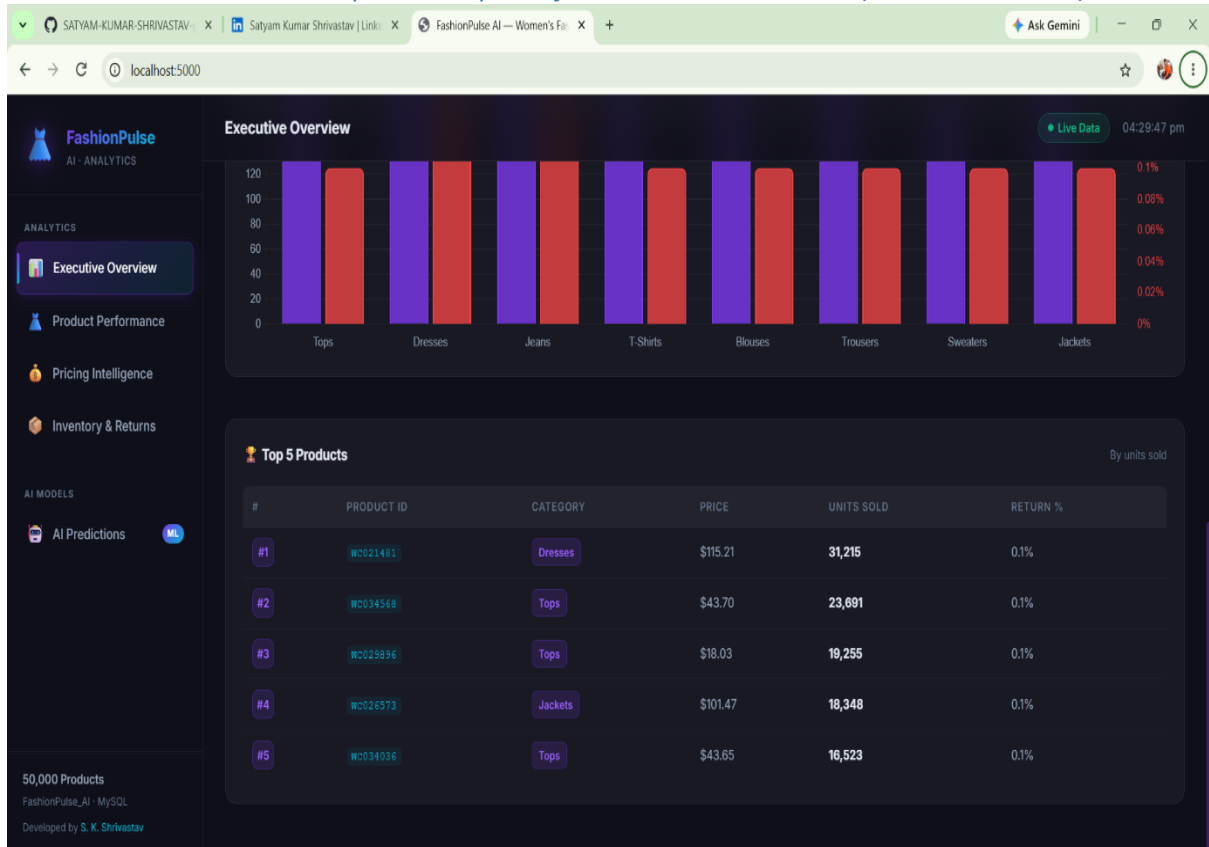


Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)



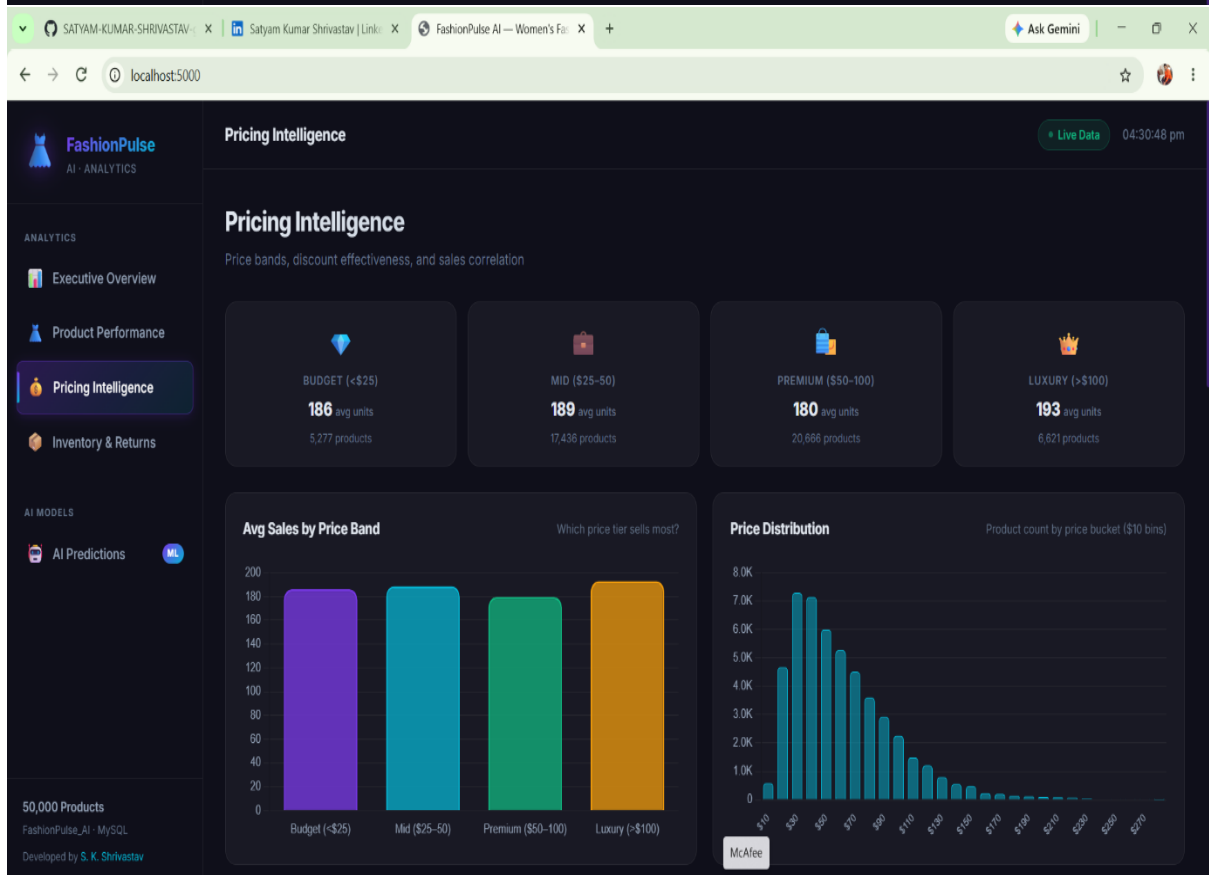
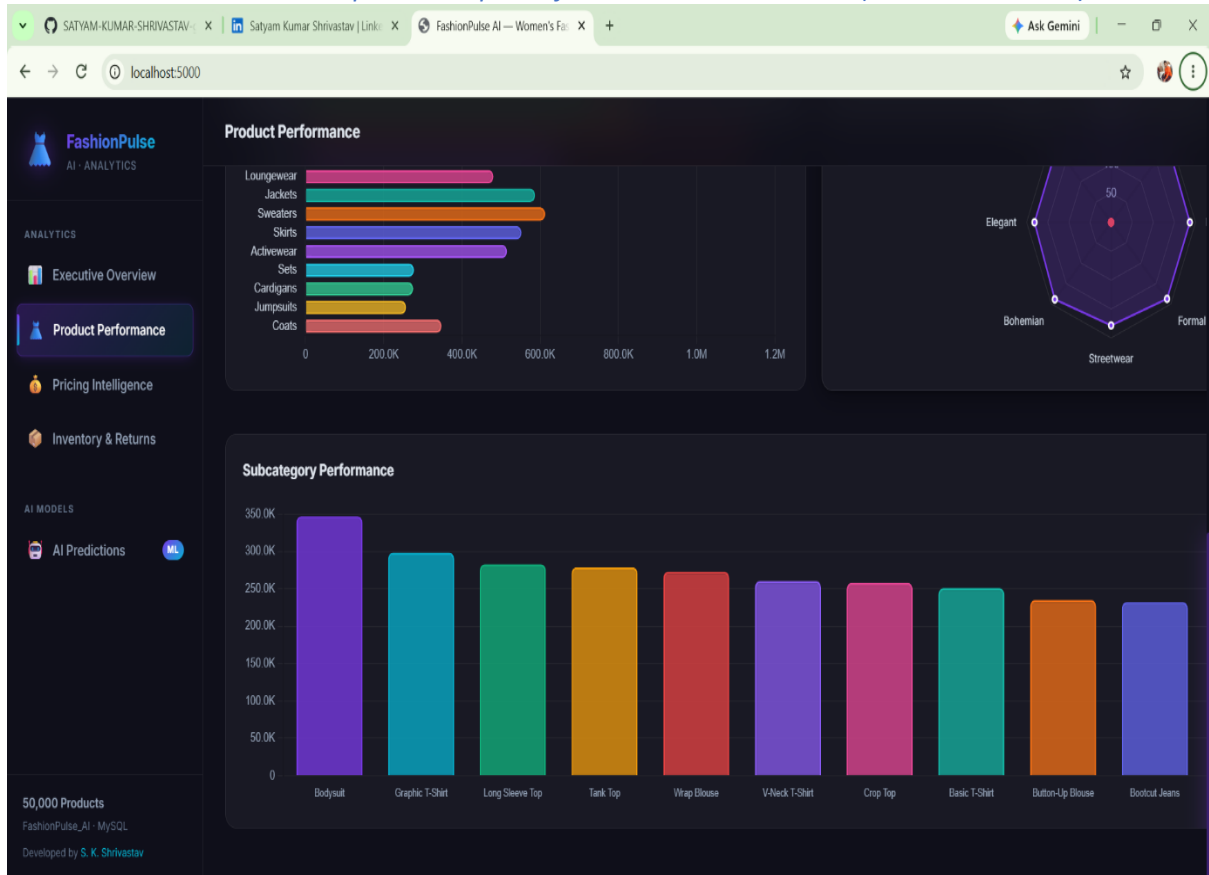


Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)



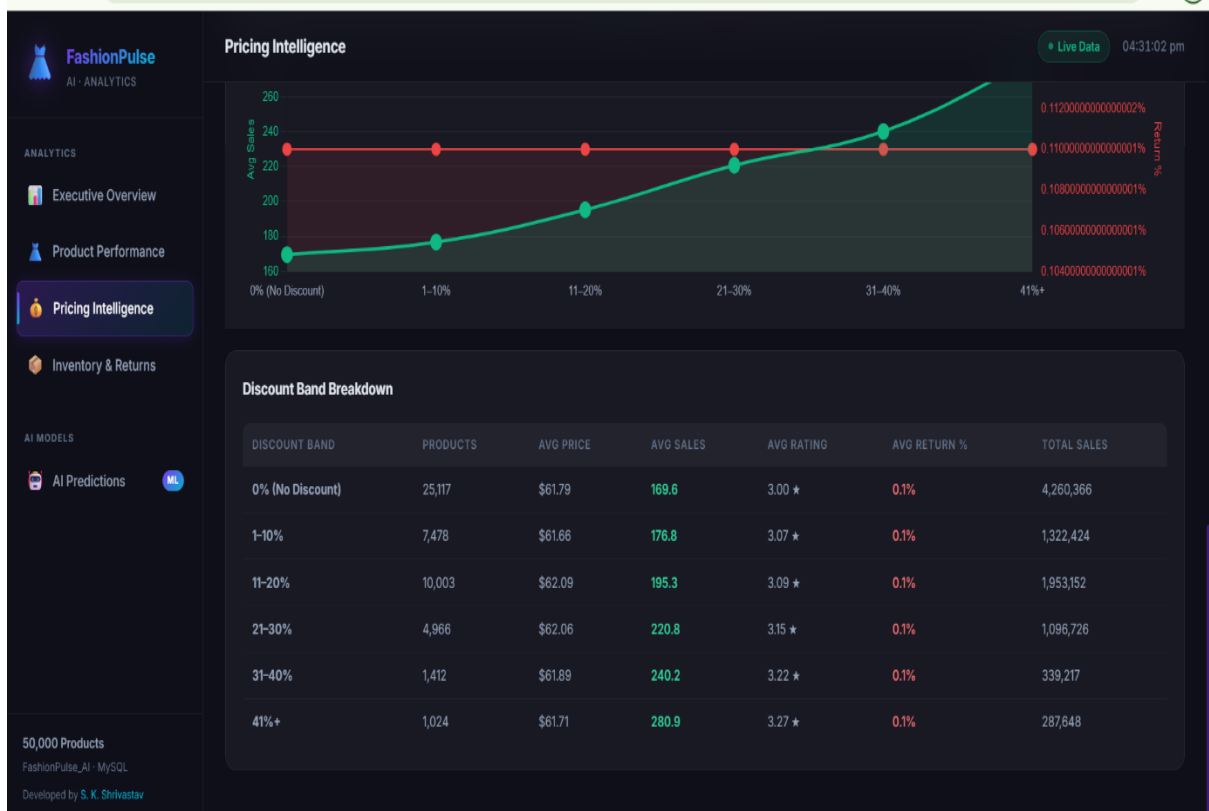
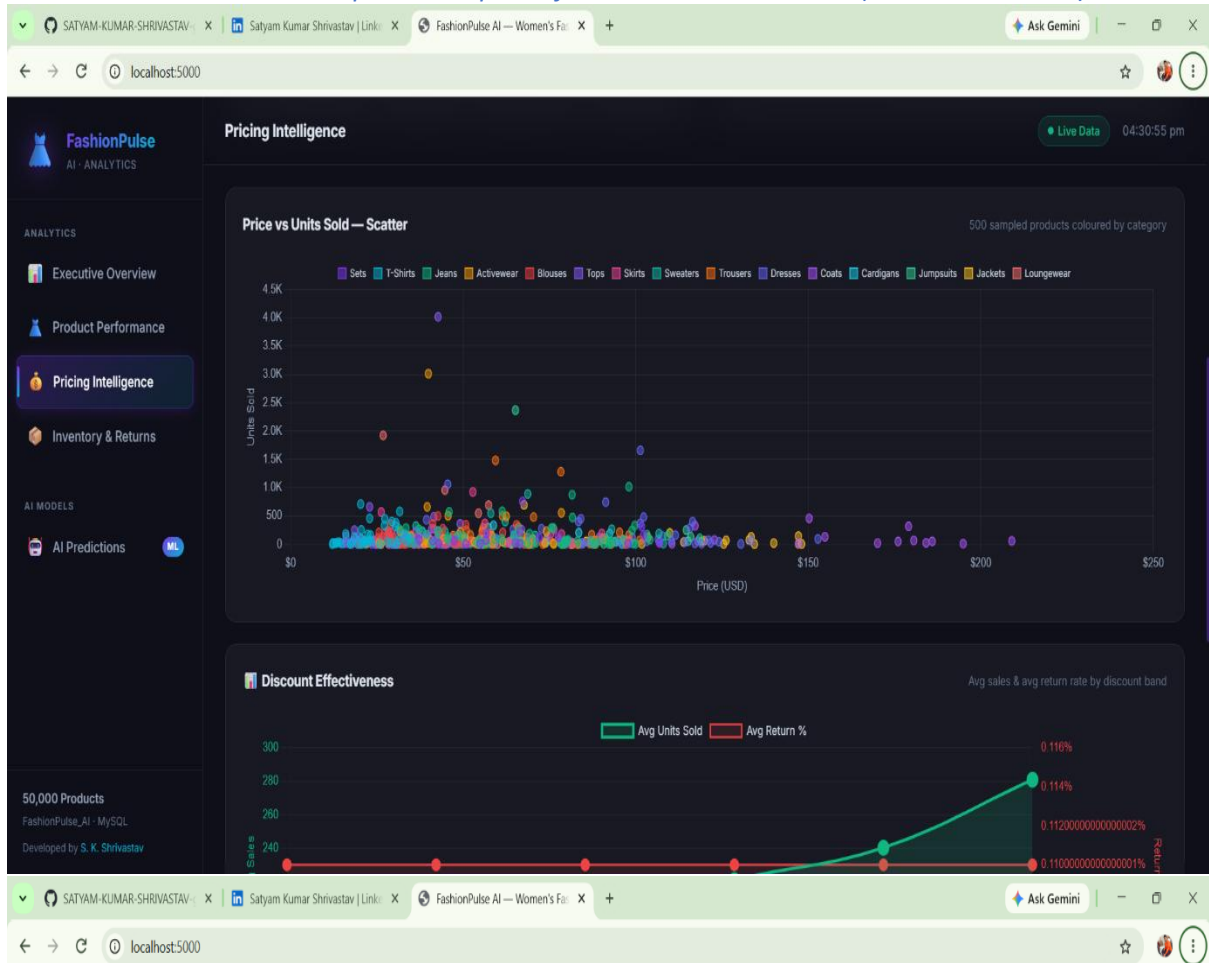


Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)



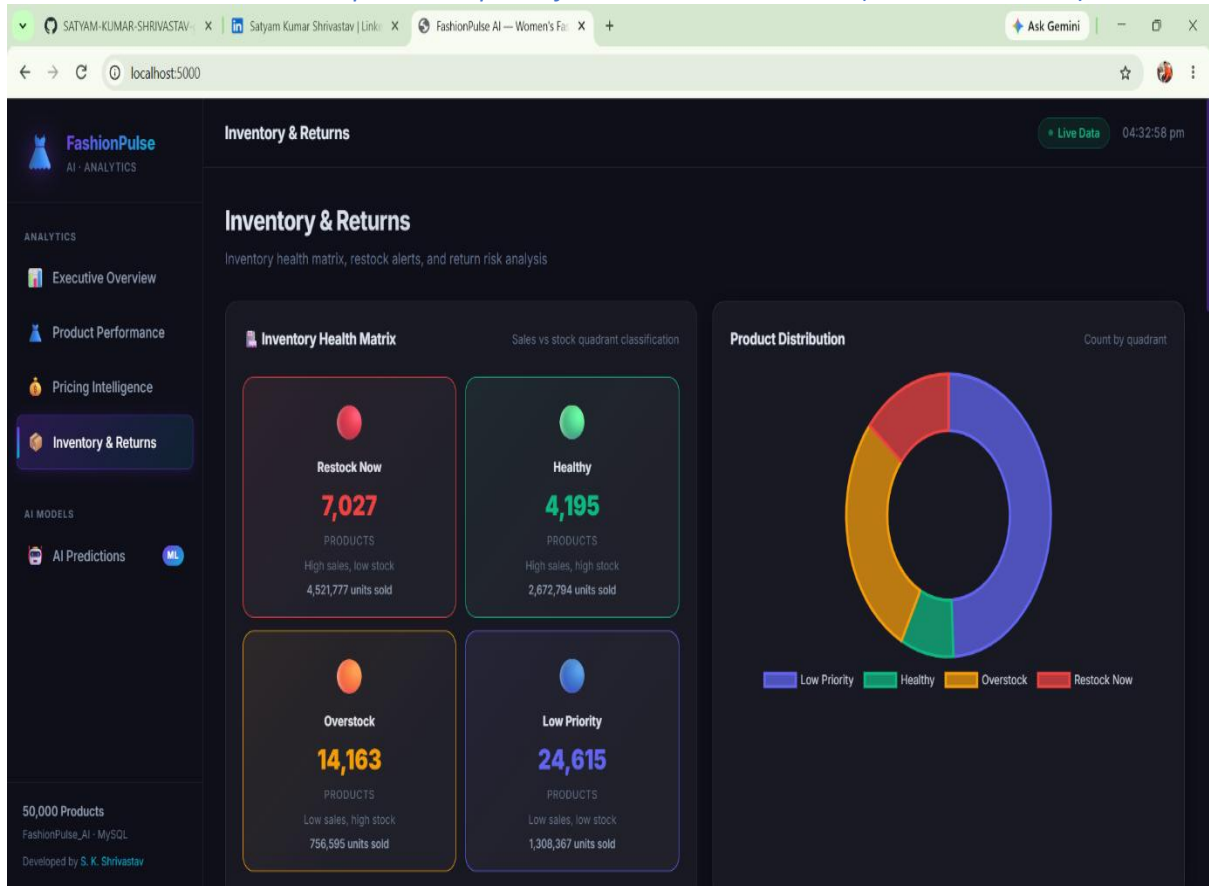


Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)





Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)



SATYAM-KUMAR-SHRIVASTAV | Satyam Kumar Shrivastav | Link | FashionPulse AI — Women's Fa | Ask Gemini | localhost:5000

FashionPulse AI - ANALYTICS

Inventory & Returns

Inventory health matrix, restock alerts, and return risk analysis

Restock Alerts

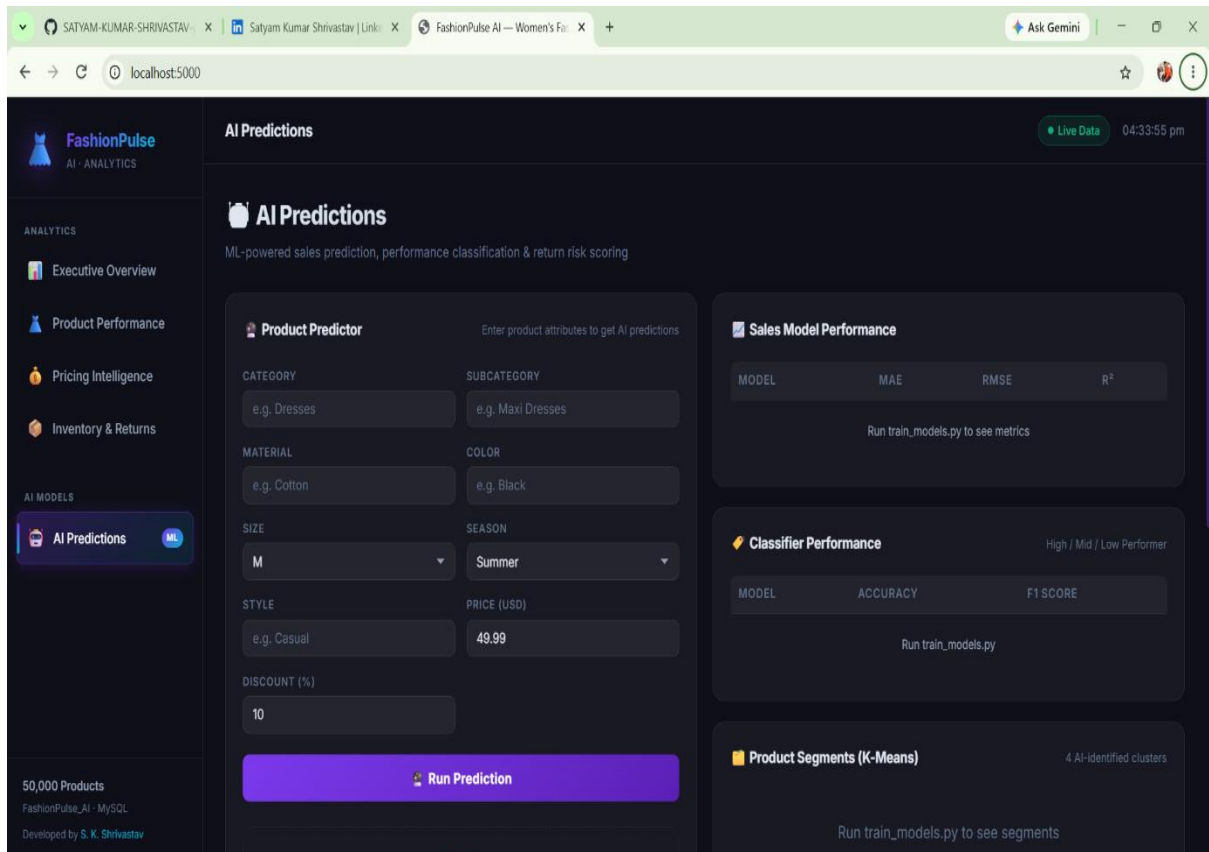
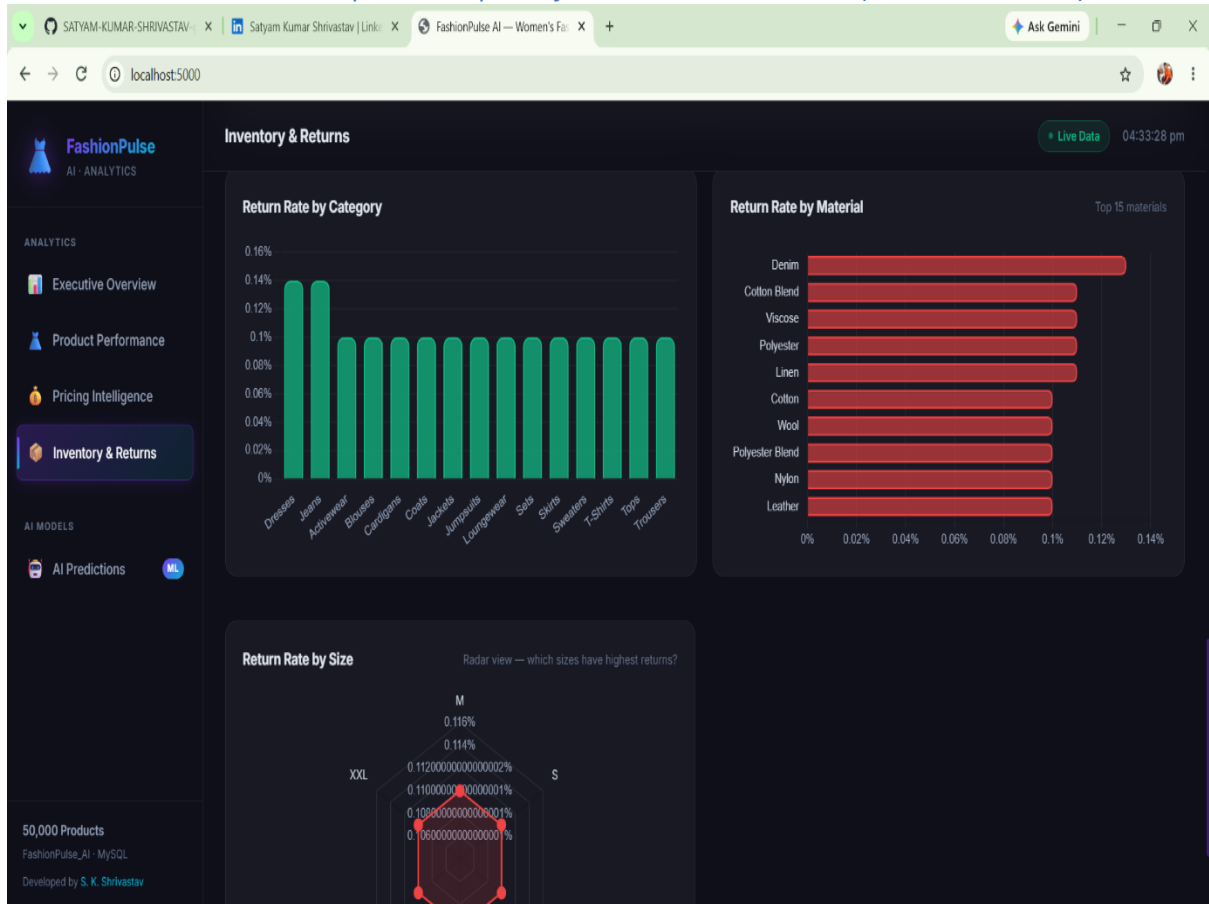
High-demand products with low inventory

PRODUCT ID	CATEGORY	SUBCATEGORY	UNITS SOLD	STOCK LEFT	RETURN %	ACTION
WC034568	Tops	Long Sleeve Top	23,691	4	0.1%	Restock
WC029896	Tops	Bodysuit	19,255	0	0.1%	Restock
WC026573	Jackets	Bomber Jacket	18,348	30	0.1%	Restock
WC034836	Tops	Bodysuit	16,523	4	0.1%	Restock
WC066413	Trousers	Wide-Leg Trousers	15,949	0	0.1%	Restock
WC038382	Sweaters	Turtleneck	14,269	0	0.1%	Restock
WC048441	Trousers	Wide-Leg Trousers	14,142	15	0.1%	Restock
WC04278	Skirts	Maxi Skirt	13,387	4	0.1%	Restock
WC026247	Trousers	Tailored Trousers	12,040	32	0.2%	Restock
WC014015	Blouses	Wrap Blouse	11,723	23	0.1%	Restock

50,000 Products
FashionPulse_AI - MySQL
Developed by S. K. Shrivastav



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)





4. Methodology and Results

4.1 Methodology

The project follows a layered workflow in which the source CSV dataset is loaded into MySQL, queried through a Flask application, visualized in a browser dashboard, and used as the training source for predictive models.

4.1.1 Data Understanding and Preparation

The dataset contains 50,000 product rows. The normalized schema includes product identifier, category, subcategory, material, color, size, season, style, price in USD, discount percentage, customer rating, review count, stock quantity, units sold, and return rate.

4.1.2 Database Design and Loading

The project uses MySQL with a database named FashionPulse_AI and a products table. The setup script creates the database and table, imports the CSV, normalizes columns, converts numeric values, refreshes existing product rows, inserts records in batches of 5,000, and performs verification checks.

4.1.3 Business Intelligence and SQL Analytics

The Flask application provides API endpoints for KPI summaries, sales by category and season, top products, category performance, style performance, price analysis, discount analysis, inventory health, and return analysis.

4.1.4 Dashboard Development

The dashboard is implemented as a single-page web application using HTML, CSS, vanilla JavaScript, and Chart.js. It contains Executive Overview, Product Performance, Pricing Intelligence, Inventory and Returns, and AI Predictions views.



4.1.5 Machine Learning Development

The predictive layer contains four workflows. The sales predictor estimates units sold using a log-transformed target and compares regression algorithms. The performance classifier assigns High Performer, Mid Performer, or Low Performer labels using sales tertiles. The return-risk predictor defines high risk using the 75th percentile of return rate. The product segmenter applies K-Means clustering to standardized product and performance characteristics.

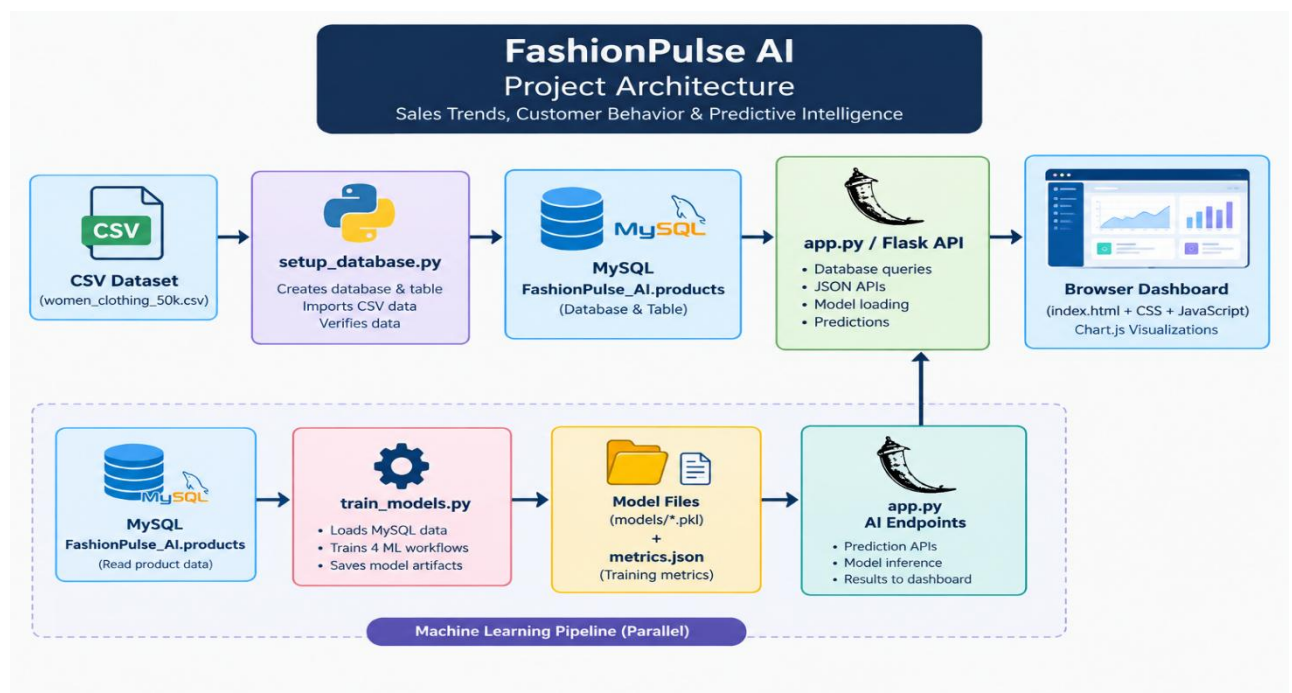
4.1.6 Model Selection and Leakage Control

Sales regression uses holdout R-squared for model selection; performance classification uses weighted F1; return-risk classification uses F1. The sales and classification feature set avoids review_count, stock_quantity, and post-sale values so predictions remain closer to information available before the sales outcome is known.

4.2 Tools and Technologies Used

Python; Flask; Flask-CORS; MySQL; MySQL Connector; SQLAlchemy; PyMySQL; Pandas; NumPy; scikit-learn; joblib; optional XGBoost and LightGBM; SciPy; HTML; CSS; vanilla JavaScript; and Chart.js.

4.3 Project Architecture





4.4 Results

The documented implementation provides KPI coverage for products, sales, estimated value, price, rating, discount, inventory, returns, and category metrics. It supports sales analysis by category and season, product/category/style/subcategory performance analysis, price and discount analysis, inventory health and restock alerts, return analysis, and model-backed predictions once training artifacts are available.

4.5 Estimated Sales Value

The dashboard uses estimated sales value = price_usd × units_sold. This is an analytical estimate rather than verified accounting revenue and does not account for refunds, taxes, shipping, cost of goods, payment fees, or currency conversion.

4.6 REST API and Application Operation

The Flask API provides /api/health, business analytics endpoints such as /api/kpis and /api/sales-by-category, and predictive endpoints such as /api/predict and /api/segment. The application runs locally at <http://localhost:5000>. Model endpoints become available after train_models.py creates the required artifacts.



5. Social/Industry Relevance of the Project

5.1 Industry Relevance

FashionPulse AI is relevant to e-commerce analytics, retail business intelligence, merchandising, inventory management, product analytics, and applied machine learning.

5.2 Practical Applications

The platform can support category performance analysis, product ranking, pricing and discount review, inventory monitoring, return-risk investigation, product segmentation, and early-stage assessment of new product attributes.

5.3 Business Value

The main business value is the conversion of a product catalog into a decision-support interface. Users can move from high-level KPIs to category, style, pricing, inventory, return, and prediction views without manually querying the database.

5.4 Industry Skills Demonstrated

The project demonstrates data preparation, relational database design, SQL analytics, REST API development, dashboard development, JavaScript visualization, machine-learning model training, model evaluation, feature selection, model persistence, application testing, and technical documentation.

5.5 Limitations

The current dataset is product-level rather than a full customer transaction history, so customer behavior is represented indirectly through product ratings, reviews, sales, inventory, and return information. The dataset has no date column, so true time-series forecasting is not implemented. Estimated sales value is not accounting revenue. The development Flask server is intended for local use rather than production deployment.



6. Learning and Reflection

6.1 Technical Learning

The project provided practical experience in connecting data engineering with business analytics and machine learning. MySQL reinforced structured data storage and repeatable loading. Flask demonstrated how analytical logic can be exposed through APIs, while JavaScript and Chart.js showed how analytical results can become an interactive interface.

6.2 Project Experience and Reflection

A key learning from FashionPulse AI is that a useful analytics project requires more than model training. Data must be stored reliably, business questions must be translated into measurable metrics, results must be exposed through usable interfaces, and predictive outputs must be presented in an interpretable way.

6.3 Overall Learning Outcome

The project strengthened practical understanding of the end-to-end analytics lifecycle: dataset ingestion, database management, SQL analysis, API development, visualization, machine learning, evaluation, and application integration.



7. Conclusion and Future Scope

7.1 Conclusion

FashionPulse AI brings together business intelligence and predictive analytics for a women's fashion e-commerce catalog. The system stores the supplied 50,000-row dataset in MySQL, exposes analytical results through Flask APIs, presents them through an interactive dashboard, and provides four machine-learning workflows for predictive sales, performance classification, return risk, and product segmentation.

7.2 Future Scope

Future development can add dated transaction-level data for genuine time-series forecasting; customer-level purchase history for deeper customer segmentation; richer campaign and seasonality features; automated retraining and model monitoring; model explainability; additional dashboard alerts; production deployment; secure environment-based database credentials; and a more advanced AI or RAG-based business knowledge layer.

7.3 Final Statement

FashionPulse AI demonstrates a practical approach to combining relational data management, business intelligence, interactive visualization, and machine learning in a single application. The project establishes a strong base for extending fashion e-commerce analytics into customer intelligence, forecasting, optimization, and production decision-support systems.



8. Project Structure and Technical Reference

8.1 Core Files

app.py, config.py, setup_database.py, train_models.py, requirements.txt, run.bat, data/women_clothing_50k.csv, models/, static/, and templates/index.html.

8.2 Model Artifacts

sales_model.pkl, classifier_model.pkl, classifier_label_encoder.pkl, return_risk_model.pkl, segmenter.pkl, and metrics.json.

8.3 Environment Requirements

Python 3.10 or newer is recommended. MySQL Server 8.x or compatible MySQL is required. The README lists Flask, Flask-CORS, MySQL Connector, SQLAlchemy, PyMySQL, pandas, NumPy, scikit-learn, joblib, XGBoost, LightGBM, SciPy, and cryptography.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)